

Problem 1:

Let R be a Dedekind Domain. Let $I \subset R$ be a nonzero ideal. Prove that $I = \mathfrak{p}_1^{e_1} \cap \cdots \cap \mathfrak{p}_r^{e_r}$ is a minimal primary decomposition of I .

Proof.

Since $\mathfrak{p}_i$ are all distinct maximal ideals this implies that $\sqrt{\mathfrak{p}_i^{e_i}} = \mathfrak{p}_i$ are all distinct. Thus, we have all unique primes. Also, these are $\mathfrak{p}_i$ -primary since they are powers of prime ideals.

First $\mathfrak{p}_i + \mathfrak{p}_j = R$ for $i \neq j$ since $\mathfrak{p}_i, \mathfrak{p}_j$ are distinct maximal primes. Hence,

$$\mathfrak{p}_i + \mathfrak{p}_j = R$$

Consider $\sqrt{\mathfrak{p}_i^{e_i} + \mathfrak{p}_j^{e_j}} = \sqrt{\mathfrak{p}_i + \mathfrak{p}_j} = R \implies \mathfrak{p}_i^{e_i} + \mathfrak{p}_j^{e_j} = R$.

Therefore, by the Chinese remainder theorem we have $\mathfrak{p}_i^{e_i} \cap \mathfrak{p}_j^{e_j} = \mathfrak{p}_i^{e_i} \mathfrak{p}_j^{e_j}$ for $i \neq j$.

Let $J_k = \bigcap_{i \neq k} \mathfrak{p}_i^{e_i} = \prod_{i \neq k} \mathfrak{p}_i^{e_i}$.

If $I = J_k$, then the unique prime-power factorization of ideals would have to agree. But in the factorization of I , the exponent on $\mathfrak{p}_k$, $e_k > 0$, whereas in J_k the exponent on $\mathfrak{p}_k$ is zero. Thus, $I \neq J_k$. Therefore, the intersection is not redundant.

Thus, this is a minimal primary decomposition of I . ■

Problem 2:**(a)****Proof.**

Since $Z = V(f)$ is irreducible this implies that $I(Z) = \sqrt{(f)}$ is prime

By assumption $\text{codim} Z = 1$ thus, $\text{codim} \sqrt{(f)} = 1$.

Let $\mathfrak{p}$ any codimension-1 prime s.t. $f \in \mathfrak{p}$. Thus, $(f) \subset \mathfrak{p}$, so $V(\mathfrak{p}) \subset V(f) = Z$.

Now take radicals and we get that $I(Z) \subset \mathfrak{p}$, but since we know that $\text{codim} \mathfrak{p} = 1$, but we are in a domain thus $(0) \subsetneq I(Z)$ is a valid chain. But this implies that $\mathfrak{p} = I(Z)$ since $\mathfrak{p}$ has codimension 1.

Thus, $I(Z)$ is the unique codimension 1 prime containing f .

Assume $\mathfrak{p}$ is a codimension 1 prime s.t. $f \notin \mathfrak{p}$.

Since prime ideals of R_f are in bijection with prime ideals $\mathfrak{q}$ that do not meet f this implies that since $f \notin \mathfrak{p}$ all the primes below $\mathfrak{p}$ are also in R_f .

Thus, $\text{codim}(\mathfrak{p}R_f) = \text{codim}(\mathfrak{p})$. ■

(b)**Proof.**

The fraction field of R_f is the same K .

Let $g \in K^\times$

In R : $\text{div}_R(g) = \sum_{(0) \neq \mathfrak{p} \neq I(Z)} v_{\mathfrak{p}}(g)[\mathfrak{p}]$

In R_f : $\text{div}_R(g) = \sum_{(0) \neq \mathfrak{q}} v_{\mathfrak{q}}(g)[\mathfrak{q}]$

Now for codim-1 primes $\mathfrak{q} \subset R_f$ are exactly $\mathfrak{q} = \mathfrak{p}R_f$ s.t. $\mathfrak{p} \neq I(Z)$ and $f \notin \mathfrak{p}$.

For such $\mathfrak{p}$ localizing shows $(R_f)_{\mathfrak{q}} \cong R_{\mathfrak{p}}$ this is true since inverting f then $R_f \setminus \mathfrak{q}$ is the same thing as inverting $R \setminus \mathfrak{p}$ since f is already in $R \setminus \mathfrak{p}$.

This implies the valuations agree so $v_{\mathfrak{q}} = v_{\mathfrak{p}}$.

Thus, $\text{div}_{R_f}(g) = \sum_{\mathfrak{p} \neq 0} v_{\mathfrak{p}}(g)[\mathfrak{p}R_f]$.

Let $\Phi : \text{Div}(R) \rightarrow \text{Div}(R_f)$ be the map defined by (a).

On the other hand we have $\Phi(\text{div}_R(g)) = \Phi(\sum_{\mathfrak{p} \neq 0} v_{\mathfrak{p}}(g)[\mathfrak{p}]) = \sum_{\mathfrak{p} \neq 0} v_{\mathfrak{p}}(g)[\mathfrak{p}R_f]$.

Thus, we get $\phi(\text{div}_R(g)) = \text{div}_{R_f}(g)$.

This implies that the image on principal divisors of R is a subset of principal divisors of R_f .

Now using the natural projection maps $\pi_R : \text{Div}(R) \rightarrow \text{CL}(R)$, $\pi_{R_f} : \text{Div}(R_f) \rightarrow \text{CL}(R_f)$

We can define $\Psi = \pi_{R_f} \circ \Phi : \text{Div}(R) \rightarrow \text{Cl}(R_f)$.

Since the principal divisors fact I showed above this implies that $\Psi(\text{div}_R(g)) = 0$ for all $g \in K^\times$.

Thus using the universal property of quotients we can factor through $\text{Cl}(R)$.

Thus, we get an induced map $\bar{\phi} : \text{Cl}(R) \rightarrow \text{Cl}(R_f)$ s.t. $\Psi = \bar{\phi} \circ \pi_R$. ■

(c)

Proof.

Consider $\mathbb{Z} \rightarrow \text{Cl}(R) \rightarrow \text{Cl}(R_f) \rightarrow 0$

Let $\bar{\phi}$ be the map from (b). Let Φ be the map defined by (a).

Let $D' \in \text{Div}(R_f)$. Then $D' = \sum_q n_q [q]$

where q is height-1 primes of R_f .

Each q is of the form $q = \mathfrak{p}R_f$ for some unique height 1 prime $\mathfrak{p} \subset R$ with $f \notin \mathfrak{p}$.

Define $D = \sum_q n_q [\mathfrak{p}] \in \text{Div}(R)$.

By definition $\Phi(D) = \sum_q n_q [\mathfrak{p}R_f] = D'$. Thus, ϕ is surjective.

Let $[D'] \in \text{Cl}(R_f)$, we have that there exists a $D \in \text{Div}(R)$ with $\Phi(D) = D'$.

Thus, $\bar{\phi}(\pi_R([D])) = \pi_{R_f}(\phi(D)) = [D']$.

Thus, this implies that $\bar{\phi}$ is surjective. This proves exactness at $\text{Cl}(R_f)$.

Now we will look at $\ker(\bar{\phi})$.

Let $[D] \in \ker(\bar{\phi})$.

Let $D = \sum_{\text{codim } \mathfrak{p}=1} n_{\mathfrak{p}} [\mathfrak{p}] \in \text{Div}(R)$.

Since $[D] \in \ker(\bar{\phi})$ this implies that $\Phi(D)$ is principal in R_f so there exists a $g \in K^\times$ s.t. $\phi(D) = \text{div}_{R_f}(g)$.

Thus, $\phi(D) = \sum_{0 \neq \mathfrak{p} \neq I(Z)} n_{\mathfrak{p}} [\mathfrak{p}R_f] = \sum_{0 \neq \mathfrak{p} \neq I(Z)} v_{\mathfrak{p}}(g) [\mathfrak{p}R_f]$.

This implies that $n_{\mathfrak{p}} = v_{\mathfrak{p}}(g)$.

Now consider the divisor of g in R

We get $\text{div}_R(g) = \sum_{\mathfrak{p} \neq 0} v_{\mathfrak{p}}(g) [\mathfrak{p}] = v_{I(Z)} [I(Z)] + \sum_{I(Z) \neq \mathfrak{p} \neq 0} n_{\mathfrak{p}} [\mathfrak{p}]$.

Now comparing D and $\text{div}_R(g)$ we get

$D - \text{div}_R(g) = (n_{I(Z)} - v_{I(Z)}) [I(Z)]$.

Then passing to class groups we get $[D - 0] = [D - \text{div}_R(g)] = (n_{I(Z)} - v_{I(Z)}) [I(Z)] \in \text{Cl}(R)$.

Thus every element of $\ker(\bar{\phi})$ is a multiple of $[I(Z)]$.

Thus, $\ker(\bar{\phi}) \subset \langle [I(Z)] \rangle$.

Also since $\Phi([I(Z)]) = 0 \implies \bar{\phi}([I(Z)]) = 0$.

Thus, $\ker(\bar{\phi}) = \langle [I(Z)] \rangle$.

Thus, define $\psi : \mathbb{Z} \rightarrow \text{Cl}(R)$ s.t. $\psi(1) = [I(Z)]$.

Thus, $\text{Im}(\psi) = \langle [I(Z)] \rangle = \ker(\bar{\phi})$.

Therefore, this is an exact sequence. ■

Problem 3:

Let $R = k[x, y, z]/(xy - z^2)$ where k is a field.

(a)

Prove that R is a normal noetherian domain.

Proof.

First R is noetherian since it is a quotient of a noetherian ring $k[x, y, z]$.

Proven in a previous class we have that $R_{\bar{x}} \cong R[t]/(xt-1) \cong k[x, y, z, t]/(xy-z^2, xt-1)$.

Consider $\phi : k[x, x^{-1}, t] \rightarrow k[x, y, z, t]/(xy - z^2, xt - 1)$ s.t. $\phi(x) = \bar{x}$, $\phi(x^{-1}) = \bar{t}$, $\phi(t) = \bar{z}\bar{t}$, and $\phi(a) = \bar{a}$ for $a \in k$.

Homomorphism:

Let $f, g \in k[x, x^{-1}, t]$ $\phi(f + g) = f(\bar{x}, \bar{t}, \bar{z}\bar{t}) + g(\bar{x}, \bar{t}, \bar{z}\bar{t}) = \phi(f) + \phi(g)$.

$\phi(fg) = f(\bar{x}, \bar{t}, \bar{z}\bar{t})g(\bar{x}, \bar{t}, \bar{z}\bar{t}) = \phi(f)\phi(g)$.

$\phi(1) = 1$.

Now we have to check the relations.

$\phi(xx^{-1}) = \bar{x}\bar{t} = 1$

Surjective:

We already have generators $\bar{x}, \bar{t}$. Just have to get $\bar{z}$ and $\bar{y}$.

$\phi(xt) = \bar{x}\bar{z}\bar{t} = \bar{z}$.

$\phi(xt^2) = \bar{x}\bar{z}^2\bar{t}^2 = \bar{z}^2\bar{t} = \bar{y}$.

Thus surjective.

Injective:

Let $f \in \ker \phi$

$\phi(f(x, x^{-1}, t)) = f(\bar{x}, \bar{t}, \bar{z}\bar{t}) = 0$.

Since $\bar{x}, \bar{z}$ are algebraically independent of one another this implies the only way for $f(\bar{x}, \bar{t}, \bar{z}\bar{t}) = 0$ is if $f = 0$.

Thus, injective.

Therefore, $R_{\bar{x}} \cong k[x, x^{-1}, t]$.

Thus, since $k[x, x^{-1}, t] \cong k[x, t]_x$, and $k[x, t]$ is a UFD this implies that $k[x, x^{-1}, t]$ is also a UFD since it is a localization of a UFD, and we don't invert 0.

Thus, $R_{\bar{x}}$ is normal since it is a UFD. Do a similar proof for $R_{\bar{y}}$.

Now consider $R_{\bar{x}} \cap R_{\bar{y}}$.

Let $f \in R_{\bar{x}} \cap R_{\bar{y}} \subset k(x, z)$.

Then $f = \sum_{ij} c_{ij} x^i z^j$

Since $f \in R_{\bar{x}}$ then $x^m f \in R$ for $m \in \mathbb{N}$. this means all have non-negative exponents which forces $j \geq 0$.

Since $f \in R_{\bar{y}}$ then $y^n f \in R$ so $z^{2n} x^{-n} f \in R$. This means that since we are multiply by x^{-n} that $i \geq n$.

Thus, $f \in R$.

Thus, $R_x \cap R_y = R$

Now let f be integral over R this implies it is integral over it is integral over $R_{\bar{x}} \cap R_{\bar{y}}$.

Thus it satisfies a monic polynomial with coefficients in $R_{\bar{x}} \cap R_{\bar{y}}$. Thus all the coefficients are also in $R_{\bar{x}}$ and $R_{\bar{y}}$.

So f is integral over $R_{\bar{x}}$ and $R_{\bar{y}}$. This implies that $f \in R_{\bar{x}}$ and $f \in R_{\bar{y}}$. Thus, $f \in R$.

Therefore, R is normal. ■

(b)

Let $\mathfrak{p} = (\bar{x}, \bar{z})$ be the ideal generated by the images of x and z in R . Prove that $\mathfrak{p}$ is not a principal ideal.

Proof.

Assume for contradiction that $\mathfrak{p}$ is principal s.t. $\mathfrak{p} = (f)$ for some $f \in R$.

Let $R' = R/\mathfrak{m}^2$ and $\mathfrak{m}' = \mathfrak{m}/\mathfrak{m}^2$

Consider $R'/\mathfrak{m}' \cong R/\mathfrak{m} \cong k[x, y, z]/(xy - z^2, x, y, z) \cong k$

Thus, since we have the injection from $k \rightarrow R'$ and the surjection from $\pi : R' \rightarrow k$ with kernel $\mathfrak{m}'$.

This implies that for $r \in R'$ consider $a = \pi(r) \in k$.

Since $\pi(r - a) = \pi(r) - \pi(a) = a - a = 0$. This implies that $r - a \in \mathfrak{m}'$.

Thus, all elements in R' can be written as $r = a + (r - a)$ for $a \in k$ and $r - a \in \mathfrak{m}'$.

Let f' be the image of f in R' , and $\mathfrak{p}'$ be the image of $\mathfrak{p}$ in R' .

Since $\mathfrak{p} \subset \mathfrak{m}$ and $\mathfrak{p} \not\subset \mathfrak{m}^2 = (\bar{x}^2, \bar{y}^2, \bar{z}^2, \bar{x}\bar{y}, \bar{x}\bar{z}, \bar{y}\bar{z})$.

We have that $f' \in \mathfrak{m}' \setminus \{0\}$.

Now for $r \in R'$ consider $rf' = (a + u)f' + af' + uf'$ s.t. $a \in k$ and $u \in \mathfrak{m}'$.

But since $u \in \mathfrak{m}'$ and $f' \in \mathfrak{m}'$ this means $uf' \in \mathfrak{m}'^2 = 0$ this implies $uf' = 0$.

Thus, $rf' = af'$. Thus, $\mathfrak{p} = (f') = k \cdot f'$.

This implies $\mathfrak{p}'$ is a 1 dimensional k -vector space.

Now $R' \cong k[x, y, z]/(xy - z^2, (x, y, z)^2) \cong k[x, y, z]/\mathfrak{m}^2$.

This implies that $R' = k \oplus kx \oplus ky \oplus kz$.

Since x, z are linearly independent this implies their span is a 2 dimensional subspace.

But since $\mathfrak{p}'$ is spanned by x, z this implies it is two dimensional as a vector space.

Thus, this is a contradiction. Thus, $\mathfrak{p}$ is not a principal ideal. ■

(c)

Prove that $\text{Cl}(R) \cong \mathbb{Z}/2$, generated by the class $[p]$ of the prime ideal from (b).

Proof.

Let $f = \bar{x} \in R$.

Then $Z = V(\bar{x})$. Consider $(\bar{x})$.

This corresponds to $(x, xy - z^2)$ in $k[x, y, z]$.

Thus, $\sqrt{(x, xy - z^2)} = \sqrt{(x, z^2)} = (x, z)$. Thus, passing it down we get that $\sqrt{(\bar{x})} = (\bar{x}, \bar{z}) = p$.

Thus from 2(a) it is the unique codimension-1 prime containing $\bar{x}$.

So problem 1(c) gives an exact sequence

$$\mathbb{Z} \rightarrow \text{Cl}(R) \rightarrow \text{Cl}(R_{\bar{x}}) \rightarrow 0$$

where $\mathbb{Z} \rightarrow \text{Cl}(R)$ sends $1 \rightarrow [p]$

Since we know that $\text{Cl}(R_{\bar{x}})$ is a UFD we know that $\text{Cl}(R_{\bar{x}}) = 0$.

Thus, we get an exact sequence

$$\mathbb{Z} \rightarrow \text{Cl}(R) \rightarrow 0$$

This implies the map $\mathbb{Z} \rightarrow \text{Cl}(R)$ is surjective.

Thus, we know that $\text{Cl}(R)$ is generated by $[p]$.

$I(V(\bar{x})) = p$ so p is the only height-1 prime ideal that contains $\bar{x}$.

Thus for $v_q(\bar{x})$ if $q = p$ then it is greater than zero and otherwise it is 0.

Thus, $\text{div}(\bar{x}) = v_p(\bar{x})[p]$.

Consider R_p :

Since $\bar{y} \notin p$ this implies it is a unit.

Since $\bar{x} = \bar{z}^2/\bar{y} \in R_p$. This implies that $\bar{x} \in (\bar{z}^2)R_p$.

Thus, $pR_p \subset (\bar{z}^2, \bar{z})R_p = (\bar{z}R_p)$.

Thus, $\bar{z}$ generates the maximal ideal of the DVR R_p . Thus, $\bar{z}$ is the uniformizer.

Thus, $v_p(\bar{x}) = v_p(\bar{z}^2) - v_p(\bar{y}) = 2 - 0 = 2$.

Thus, $\text{div}(\bar{x}) = 2[p]$.

Since $\bar{x} \neq 0$ this implies $\bar{x} \in K^\times$.

Thus, $[\text{div}(\bar{x})] = 0 \implies 2[p] = 0$. But since we know that p is not principal we know that $[p] \neq 0$.

Therefore, $\text{Cl}(R) \cong \mathbb{Z}/2$. ■

Problem 4:

Compute the following tensor products:

Proof.

1. $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}$

Since $\mathbb{Q} \cong S^{-1}\mathbb{Z}$ this implies that $S^{-1}\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q} \cong S^{-1}\mathbb{Q} \cong \mathbb{Q}$.

2. $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z}$

$S^{-1}\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z} \cong S^{-1}(\mathbb{Q}/\mathbb{Z})$. Since $S = \mathbb{Z} \setminus \{0\}$. This implies that $S^{-1}(\mathbb{Q}/\mathbb{Z}) \cong 0$.

3. $\mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z}$.

Let $a/b \otimes_{\mathbb{Z}} c/d \in \mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z}$.

Then $a/b \otimes_{\mathbb{Z}} c/d = ad/bd \otimes_{\mathbb{Z}} c/d = a/bd \otimes_{\mathbb{Z}} cd/d = a/bd \otimes_{\mathbb{Z}} c/1 = a/bd \otimes_{\mathbb{Z}} 0 = 0(a/bd \otimes_{\mathbb{Z}} 0) = 0 \otimes_{\mathbb{Z}} 0$.

This implies every element is zero thus $\mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z} \cong 0$.

4. $\mathbb{Z}[1/2] \otimes_{\mathbb{Z}} \mathbb{Z}[1/3]$

Let $S = \{1/2^n \mid n \in \mathbb{N}\}$ then $S^{-1}\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}[1/3] \cong S^{-1}(\mathbb{Z}[1/3]) \cong \mathbb{Z}[1/6]$.

5. $\mathbb{Z}[1/2] \otimes_{\mathbb{Z}} \mathbb{Z}/3 \cong S^{-1}\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/3 \cong S^{-1}(\mathbb{Z}/3) \cong \mathbb{Z}/3$ since $\mathbb{Z}/3$ is a field, and it did not invert 0.

(a)

Prove that an R -module M is projective if and only if M is a direct summand of a free R -module

Proof.

Assume M is projective.

Let F have one basis vector for each element of M .

Thus, there exists a surjective R -linear map $\pi : F \rightarrow M$ sending the basis vectors to its associated element.

Thus consider the short exact sequence

$$0 \rightarrow \ker(\pi) \rightarrow F \rightarrow M \rightarrow 0$$

Since M is projective this implies that

$$0 \rightarrow \operatorname{Hom}_R(M, \ker(\pi)) \rightarrow \operatorname{Hom}_R(M, F) \rightarrow \operatorname{Hom}_R(M, M) \rightarrow 0$$

is also exact.

This implies $\text{Hom}_R(M, F) \rightarrow \text{Hom}_R(M, M)$ is surjective.

Let $f : M \rightarrow M$ be the identity.

Since $\text{Hom}_R(M, F) \rightarrow \text{Hom}_R(M, M)$ is surjective this implies there exists a R -linear $g : M \rightarrow F$ s.t. $\pi \circ g = f$.

Consider $s(M) + \ker(\pi)$.

Let $x \in F$. Let $m = \pi(x) \in M$.

Consider $x - s(m) \in F$.

$\pi(x - s(m)) = \pi(x) - \pi(s(m)) = \pi(x) - f(m) = m - m = 0$. Thus, $x - s(m) \in \ker(\pi)$.

Thus, $x = s(m) + (x - s(m))$ with $s(m) \in s(M)$ and $x - s(m) \in \ker(\pi)$.

Thus, $F = s(M) + \ker(\pi)$.

Consider $s(M) \cap \ker(\pi)$.

Let $x \in s(M) \cap \ker(\pi)$.

Thus, $x = s(m)$ for some $m \in M$ and $\pi(x) = 0$.

Thus, $\pi(x) = \pi(s(m)) = m = 0$.

Thus, $m = 0 \implies s(m) = 0 = x$.

Thus, $s(M) \oplus \ker(\pi) = F$.

Consider $s : M \rightarrow s(M)$.

Surjective by definition.

Injective:

Let $x \in \ker(s)$ then $s(x) = 0 \implies \pi(s(x)) = 0 \implies f(x) = 0 \implies x = 0$.

Thus, $s(M) \cong M$ therefore, $F \cong M \oplus \ker(\pi)$. Therefore, M is a direct summand of a free R -module.

Assume M is a direct summand of a free R -module.

Thus, there exists a R -module Q s.t. $M \oplus Q \cong F$ s.t. F is a free R -module.

Let A, B, C be R -modules s.t.

$0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ is a short exact sequence.

Consider $0 \rightarrow \text{Hom}_R(M, A) \xrightarrow{f^*} \text{Hom}_R(M, B) \xrightarrow{g^*} \text{Hom}_R(M, C) \rightarrow 0$.

s.t. $f^* = f \circ (-)$ and so on.

Let $\phi \in \ker(f^*)$ thus $f \circ \phi = 0$.

Since f is injective this implies that $\phi = 0$.

Thus, f^* is injective.

Let $\phi \in \text{Im}(f^*)$.

Thus, $\phi = f \circ \psi$ for some $\psi \in \text{Hom}_R(M, A)$.

Consider $g^*(\phi) = g^*(f \circ \psi) = g \circ f \circ \psi$.

Since $\text{Im}(f) = \ker(g)$ this implies that $g \circ f \circ \psi = 0$.

Thus, $\text{Im}(f^*) \subset \ker(g^*)$.

Let $\phi \in \ker(g^*)$ thus, $g \circ \phi = 0$.

Thus, $\phi(M) \subset \ker(g) = \text{Im}(f)$.

This means every element $\phi(m) \in B$ there exists an element $a_m \in A$ s.t. $f(a_m) = \phi(m)$.

Define $\psi : M \rightarrow A$ s.t. $\psi(m) = a_m$ defined above.

Well-Defined:

This is well-defined since f is injective. If $f(a_m) = f(a_{m'}) = \phi(m) \implies a_m = a_{m'}$.

Also for $m \in M$ we get $f(\psi(m)) = f(a_m) = \phi(m)$.

R -linear:

Let $r_1, r_2 \in R$ and $m_1, m_2 \in M$

Consider $f(\psi(r_1m_1 + r_2m_2)) = \phi(r_1m_1 + r_2m_2) = r_1\phi(m_1) + r_2\phi(m_2) = r_1f(\psi(m_1)) + r_2f(\psi(m_2)) = f(r_1\psi(m_1) + r_2\psi(m_2))$.

Since f is injective we get

$\psi(r_1m_1 + r_2m_2) = r_1\psi(m_1) + r_2\psi(m_2)$.

Thus, ψ is R -linear.

Thus, $f \circ \psi = f \circ \phi = 0$.

Thus, $\phi \in \text{Im}(f^*)$.

Therefore, $\text{Im}(f^*) = \ker(g^*)$.

Let $\phi \in \text{Hom}_R(M, C)$

Since $F = M \oplus Q$, define $\psi : F \rightarrow C$ s.t. $\psi(m) = \phi(m)$ and $\psi(q) = 0$ for $m \in M$ and $q \in Q$.

This is well-defined and R -linear since $F = M \oplus Q$.

Since F is free let $\{e_i \mid i \in I\}$ be a basis set for F . Now pick $b_i \in B$ s.t. $g(b_i) = \psi(e_i)$.

Now define $\varphi : F \rightarrow B$ s.t. $\varphi(e_i) = b_i$.

That's an R -linear map $g \circ \varphi = \psi$.

Now let $i : M \rightarrow F$ be the inclusion map.

Define $\Phi = \varphi \circ i : M \rightarrow B$.

Now consider $g \circ \Phi = g \circ \varphi \circ i = \psi \circ i = \phi$.

Thus, $g \circ \Phi = \phi$.

Thus, g^* is surjective.

Therefore, the sequence is short exact. ■

(b)

Prove that if M is a projective R -module, then M is flat.

Proof.

Assume M is a projective R -module.

This implies from (a) it is a direct summand of a free module.

Thus, there exists a R -module N s.t. $M \oplus N = F$ s.t. F is free.

Thus, we know that $- \otimes_R F$ is exact since F is free.

Since F is a direct sum this implies that $- \otimes_R F \cong (- \otimes_R M) \oplus (- \otimes_R N)$.

Let $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ be a short exact sequence.

Since F is free this implies that

$0 \rightarrow A \otimes F \rightarrow B \otimes F \rightarrow C \otimes F \rightarrow 0$ is exact.

Thus, $0 \rightarrow (A \otimes M) \oplus (A \otimes N) \rightarrow (B \otimes M) \oplus (B \otimes N) \rightarrow (C \otimes M) \oplus (C \otimes N) \rightarrow 0$ is also exact.

Consider $0 \rightarrow (A \otimes M) \rightarrow (B \otimes M) \rightarrow (C \otimes M) \rightarrow 0$

The only difference with these maps is you take the inclusion map from $A \otimes M \rightarrow A \otimes (A \otimes M) \oplus (A \otimes N)$ and then apply the function.

This will also give us an exact sequence.

Thus, M is also flat. ■

(c)

Prove that if M is a finitely generated projective R -module, then M is finitely presented.

Proof.

Assume M is a finitely generated projective R -module.

Since M is finitely generated this implies that there exists a surjective R -linear map $\pi : R^n \twoheadrightarrow M$.

We can do the same proof as in part (a) to show that $R^n \cong M \oplus \ker(\pi)$.

Since $\ker(\pi)$ is a direct summand of a finitely generated module this implies $\ker(\pi)$ is also finitely generated.

Thus, this implies we have a surjective R -linear map $f : R^m \twoheadrightarrow \ker(\pi)$.

Also let $i : K \rightarrow R^n$ be the inclusion map.

Now consider $R^m \xrightarrow{i \circ f} R^n \xrightarrow{\pi} M \rightarrow 0$.

Consider $\text{Im}(i \circ f) = i(\text{Im}(f)) = i(\ker(\pi)) = \ker(\pi)$.

Also π is surjective.

Thus, this sequence is exact.

Therefore, M is finitely presented. ■

Problem 5:

Let $f_1 : A \rightarrow R_1$ and $f_2 : A \rightarrow R_2$ be ring homomorphisms. Construct an explicit bijection between $\text{Spec}(R_1 \otimes_A R_2)$ and the following set:

$$\left\{ (p_1, p_2, q, r) \left| \begin{array}{l} p_i \in \text{Spec}(R_i), q \in \text{Spec}(A), \\ f_i^{-1}(p_i) = q, r \in \text{Spec}(\kappa(p_1) \otimes_{\kappa(q)} \kappa(p_2)) \end{array} \right. \right\}$$

Proof.

Call the set B .

Let $\phi_1 : R_1 \rightarrow R_1 \otimes_A R_2, r_1 \mapsto r_1 \otimes 1$. Define ϕ_2 a similar way for $R_2 \rightarrow R_1 \otimes_A R_2$.

Let $P \in \text{Spec}(R_1 \otimes_A R_2)$.

Define $p_1 = \phi_1^{-1}(P), p_2 = \phi_2^{-1}(P)$.

Define $q = f_1^{-1}(p_1)$ and $q' = f_2^{-1}(p_2)$. Shown in class we have a commutative diagram s.t. $\phi_1(f_1(a)) = f_1(a) \otimes_A 1 = 1 \otimes_A f_2(a) = \phi_2(f_2(a))$,

Thus, $q = q'$.

Let $\kappa(P)$ to be the residue field of P in $R_1 \otimes_A R_2$.

Consider the maps

$$R_i \xrightarrow{\phi_i} R_1 \otimes_A R_2 \xrightarrow{i} (R_1 \otimes_A R_2)_P \xrightarrow{\pi} \kappa(P)$$

Call this map $\psi_i : R_i \rightarrow \kappa(P)$.

Call $R_1 \otimes R_2 = S$.

Let $s \in P$. Then $s/1 \in PS_P$ thus $s/1 \in \ker(\pi)$ thus $s \in \ker(\pi \circ i)$.

Let $s \in \ker(\pi \circ i)$. Then $\pi \circ i(s) = 0 \implies s/1 \in PS_P$.

This implies $s \in P$. Thus, $\ker(\pi \circ i) = P$.

Therefore, $\ker(\psi_i) = \ker(\pi \circ i \circ \phi_i) = \phi_i^{-1}(\ker(\pi \circ i)) = p_i$.

This implies by the universal property of quotients ψ_i factors through

$$R_i \rightarrow R_i / p_i \rightarrow \kappa(P)$$

Then again by universal property of localization we get that it factors again through

$$R_i \rightarrow R_i / p_i \rightarrow \kappa(p_i) \rightarrow \kappa(P)$$

Now consider the maps ρ

$$A \xrightarrow{f_i} R_i \xrightarrow{\phi_i} \kappa(P)$$

Let $a \in m f q$, then $f_i(a) \in p_i = \ker(\psi_i)$, so $\rho(a) = \psi_i(f_i(a)) = 0$

Let $a \in \ker(\rho)$ then $\rho(a) = 0 \implies \psi_i(f_i(a)) = 0 \implies f_i(a) \in \ker(\psi_i) = \mathfrak{p}_i$. Thus, $a \in f_i^{-1}(\mathfrak{p}_i) = \mathfrak{q}$.

Thus, $\ker(\rho) = \mathfrak{q}$.

Doing the same sort of thing we get it factors through the quotient then the localization

$$A \rightarrow A/\mathfrak{q} \rightarrow \kappa(\mathfrak{q}) \rightarrow \kappa(P)$$

Call the induced map from $\kappa(\mathfrak{q}) \rightarrow \kappa(P)$, $\tilde{\rho}$.

Now look at the composite map $A \rightarrow R_i \rightarrow \kappa(\mathfrak{p}_i)$

If $a \in \mathfrak{q}$ then $f_i(a) \in \mathfrak{p}_i$, so its image in $\kappa(\mathfrak{p}_i)$ is 0. Thus, $\mathfrak{q} \subset \ker(A \rightarrow \kappa(\mathfrak{p}_i))$

If $(A \rightarrow \kappa(\mathfrak{p}_i))(a) = 0$ then $f_i(a)$'s image in $R_i/\mathfrak{p}_i$ is 0. Thus, $f_i(a) \in \mathfrak{p}_i$. Thus, $a \in f_i^{-1}(\mathfrak{p}_i) = \mathfrak{q}$.

Thus, $\ker(A \rightarrow \kappa(\mathfrak{p}_i)) = \mathfrak{q}$. Thus, again factoring it through the same two things we get that

$$A \rightarrow \kappa(\mathfrak{q}) \rightarrow \kappa(\mathfrak{p}_i).$$

Now putting it all together we get

$$A \rightarrow \kappa(\mathfrak{q}) \rightarrow \kappa(\mathfrak{p}_i) \rightarrow \kappa(P).$$

Call the induced maps $\Psi_i : \kappa(\mathfrak{q}) \rightarrow \kappa(\mathfrak{p}_i)$ and $\Phi_i : \kappa(\mathfrak{p}_i) \rightarrow \kappa(P)$.

First, the Ψ_i make $\kappa(\mathfrak{p}_i)$ $\kappa(\mathfrak{q})$ -algebras.

Now define

$$\Phi : \kappa(\mathfrak{p}_1) \times \kappa(\mathfrak{p}_2) \rightarrow \kappa(P) \text{ s.t. } \Phi(x, y) = \Phi_1(x)\Phi_2(y).$$

Bilinear:

Let $a \in \kappa(\mathfrak{q})$.

Consider $\Phi(ax, y) = \Phi_1(ax)\Phi_2(y) = \rho(\tilde{a})\Phi_1(x)\Phi_2(y)$. Since both $\Phi_1(x)$ restrict to the same $\tilde{\rho}$. This implies that

$$\rho(\tilde{a})\Phi_1(x)\Phi_2(y) = \Phi_1(x)\Phi_2(ay) = \Phi(x, ay).$$

Multiplicative + Additive:

$$\Phi(x_1 \circ x_2, y_1 \circ y_2) = \Phi_1(x_1 \circ x_2) \circ \Phi_2(y_1 \circ y_2) = \Phi_1(x_1) \circ \Phi_2(y_1) \circ \Phi_1(x_2) \circ \Phi_2(y_2) = \Phi(x_1, y_1) \circ \Phi(x_2, y_2)$$

for $\circ \in \{\times, +\}$.

Identity:

$$\Phi(1, 1) = \Phi_1(1)\Phi_2(1) = 1 \otimes 1$$

Thus, by universal property of tensor product we get a $\kappa(\mathfrak{q})$ -algebra Homomorphism

$$\tau : \kappa(\mathfrak{p}_1) \otimes_{\kappa(\mathfrak{q})} \kappa(\mathfrak{p}_2) \rightarrow \kappa(P) \text{ s.t.}$$

$$\tau(x \otimes y) = \Phi_1(x)\Phi_2(y).$$

Now define $\mathfrak{r} = \ker(\tau)$

Since $\kappa(P)$ is a domain we get that (0) is prime. Thus, $\ker(\tau)$ is prime.

Thus, $r \in \text{Spec}(\kappa(\mathfrak{p}_1) \otimes_{\kappa(\mathfrak{q})} \kappa(\mathfrak{p}_2))$.

Now define $\theta : \text{Spec}(R_1 \otimes_A R_2) \rightarrow B$ s.t $\theta(P) = (\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{q}, r)$. This is defined from all the work before.

Now given a tuple $(\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{q}, r) \in B$.

Let $k = \kappa(\mathfrak{q})$, $k_i = \kappa(\mathfrak{p}_i)$ and $B = k_1 \otimes_k k_2$.

Since $r \in \text{Spec}(B)$ let $K = \text{Frac}(B/r)$.

We have the canonical maps

$$k_i \rightarrow B \rightarrow B/r \rightarrow K$$

and the maps

$$R_i \rightarrow k_i$$

Composing we get the ring homomorphisms $\psi_i : R_i \rightarrow K$

Now again using the commutative diagram of tensor products we get that $\psi_1 \circ f_1 = \psi_2 \circ f_2 : A \rightarrow K$.

Thus, again like above define $\phi_{(\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{q}, r)} : R_1 \times R_2 \rightarrow K$

s.t. $\phi_{(\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{q}, r)}(x, y) = \psi_1(x)\psi_2(y)$.

Then doing the same proof as above to show this is A -bilinear and a ring homomorphism.

We get a map by the universal property

$$\Phi_{(\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{q}, r)} : R_1 \otimes_A R_2 \rightarrow K.$$

Define $P = \ker(\Phi_{(\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{q}, r)}) \in \text{Spec}(R_1 \otimes_A R_2)$ which is again prime since K is a domain thus the preimage of (0) which is prime is now prime.

Now define $\alpha : B \rightarrow \text{Spec}(R_1 \otimes_A R_2)$ by $\alpha((\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{q}, r)) = P$ which was shown as above.

These are inverses due to how the maps used in defining α, τ are all the same. ■